

Project Design Phase-II
Solution Requirements (Functional & Non-functional)

Date	22 July 2026
Team ID	TEAM-HT-2026
Project Name	Heritage Treasures: An In-Depth Analysis of UNESCO World Heritage Sites in Tableau
Maximum Marks	4 Marks

Functional Requirements:

Following are the functional requirements of the proposed solution.

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	Data Collection & Extraction	Collect the UNESCO World Heritage Sites (2019) dataset Connect the dataset to Tableau (Text file / Excel / Database connector)
FR-2	Data Preparation	Clean nulls and duplicates Standardize Country / Region names Create calculated fields (Danger Flag, Years Since Inscription, Region Rank)
FR-3	Data Visualization	Treemap of heritage sites by country Pie chart of sites In Danger vs Not in Danger Line chart of regional inscription trends
FR-4	Dashboard Design	Combine visualizations into one responsive dashboard Add interactive filters (Region, Category, Year range) Enable cross-filtering/highlighting between charts
FR-5	Story Creation	Build a multi-scene story: Overview → At-Risk Sites → Regional Trends → Takeaways Add captions guiding the viewer through each scene
FR-6	Web Integration & Publishing	Publish workbook to Tableau Server/Public Embed dashboard and story into a Flask web application UI

Non-functional Requirements:

Following are the non-functional requirements of the proposed solution.

FR No.	Non-Functional Requirement	Description
NFR-1	Usability	Dashboard and story must be intuitive for non-

		technical stakeholders (policy makers, tourism boards) with clear labels, legends and tooltips.
NFR-2	Security	Published workbook access is restricted via Tableau Server/Public permissions; the Flask app validates any user input before use.
NFR-3	Reliability	Visualizations must consistently reflect the underlying dataset with no broken calculated fields or missing data after refresh.
NFR-4	Performance	Dashboard should load and respond to filter interactions within a few seconds even with the full dataset loaded.
NFR-5	Availability	The embedded dashboard/story should remain accessible whenever the Flask web app and Tableau Server/Public link are online.
NFR-6	Scalability	Design should accommodate additional years of UNESCO data or extra countries/regions without redesigning the workbook.